Pennsylvania State University - American Society of Mechanical Engineers Club

Considerations of Various Hovercraft Skirt Material



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1 Research Goals

A hovercraft is composed of many parts that allows it to sit on a pocket of high-pressure air and effortlessly glide across surfaces such as water and land. One of the most important aspects of a hovercraft is the fact it is sitting on a pocket of air, to achieve this feature we need a component known as a skirt. The skirt goes around the outer perimeter of the hovercraft and allows it to trap a pocket of air. Thus, we need to take careful consideration regarding the choice of material for the skirt.

1.1 Material Property Considerations

- Durability
- Abrasion Resistance
- UV Resistance
- Weather Resistance
- Water Impermeability
- Air Impermeability
- Material Cost
- Material Weight
- Ease of Use
- Skid Resistance

1.2 Value of Material Property

While most of these parameters are important, there are some that we must prioritize over others thus we have a values system. Ranging from 1-5 solely based on user perception with bias on project needs:

- Durability: 4.1
- Abrasion Resistance: 4.5
- UV Resistance: 1.3
- Weather Resistance: 1.9
- Water Impermeability: 4.4
- Air Impermeability: 4.6
- Material Cost: 2.1
- Material Weight: 4.1
- Ease of Use: 3.4
- Skid Resistance: 3.9

Abrasion resistance, durability, water impermeability, air impermeability, and material weight are the most important factors we need to take into consideration. They are the most determining factors in the performance of our design. Material cost, Skid Resistance, and Ease of Use are secondary factors because the material cost can be partially excused for its importance so we can spend more if necessary. Ease of use is important, but we have a vast amount of tools at our disposal so we should be able to overcome any material properties if they are present. Skid Resistance is somewhat importance as it controls how much friction the hovercraft experiences as a result of the skirt dragging along the ground. UV resistance and weather resistance are tertiary factors we need to take into consideration, as our project is not designed to handle environmental factors for a significant amount of time we do not need to take drastic measures to ensure these properties are present in our material.

1.3 Physical Principle Consideration

While the physical design of the hovercraft has not been laid out one should note that one of the parameters of our material choice depends on the design choices of the hovercraft. As previously mentioned, the hovercraft is a concept designed on the principle of sitting atop a pocket of air. One should take into account that putting weight atop this pocket of air will result in the air becoming compressed and thus exerting a force on the skirt. However, if the pressure stress is spread over a larger area then we can reduce the stress on the skirt while maintaining the same force. Thus the equation describing this behavior can be seen as the following

$$t_{hickness} \propto C_{oefficient} \cdot \frac{W_{eight}}{S_{urfaceArea}}$$

Failure to take this into account may result in underperformance or a pressure blow out. However, due to the financially strained nature of the project, there is a low chance the design will functionally exceed the material strength.

2 Considered Materials

After a quick cursory gloss, we have come across the following materials

- Neoprene Rubber
- Neoprene Fabric
- Polyurethane-coated Nylon
- Hypalon Fabric
- PVC Coated Fabric
- Polyethylene Fiber
- Polyvinal Fabric
- Cordura Fabric

2.1 Neoprene Rubber

Neoprene (also polychloroprene) is a family of synthetic rubbers that are produced by polymerization of chloroprene. Neoprene exhibits good chemical stability and maintains flexibility over a wide temperature range. In this case, it is formed as a uniform rubber sheet.

2.1.1 Neoprene Rubber Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 4.6
- Abrasion Resistance: 4.6
- UV Resistance: 4.3
- Weather Resistance: 4.3
- Water Impermeability: 2.6
- Air Impermeability: 3.4
- Material Cost: 3.5
- Material Weight: 2.3
- Ease of Use: 3.8
- Skid Resistance: 1.2

2.2 Neoprene Fabric

Neoprene (also polychloroprene) is a family of synthetic rubbers that are produced by polymerization of chloroprene. Neoprene exhibits good chemical stability and maintains flexibility over a wide temperature range. In this case, it is formed as a uniform fabric sheet.

2.2.1 Neoprene Fabric Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 4.7
- Abrasion Resistance: 4.7
- UV Resistance: 4.0
- Weather Resistance: 4.5
- Water Impermeability: 4.6
- Air Impermeability: 4.6
- Material Cost: 2.8
- Material Weight: 1.9
- Ease of Use: 3.5
- Skid Resistance: 1.6

2.3 Polyurethane-coated Nylon

Polyurethane (PU) Coated Nylon is a type of fabric that is made by applying a thin layer of polyurethane onto one side of a nylon textile. This coating provides the fabric with increased strength, abrasion resistance, and water repellency.

2.3.1 Polyurethane-coated Nylon Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 4.6
- Abrasion Resistance: 4.7
- UV Resistance: 5.0
- Weather Resistance: 4.8
- Water Impermeability: 4.3
- Air Impermeability: 4.5
- Material Cost: 4.8
- Material Weight: 4.8
- Ease of Use: 4.6
- Skid Resistance: 4.3

2.4 Hypalon Fabric

This is a synthetic rubber coating onto polyester or nylon fabric usually with a neoprene coating on the back of the fabric.

2.4.1 Hypalon Fabric Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 4.5
- Abrasion Resistance: 4.6
- UV Resistance: 4.3
- Weather Resistance: 4.4
- Water Impermeability: 5.0
- Air Impermeability: 4.3
- Material Cost: 3.3
- Material Weight: 3.2
- Ease of Use: 4.7
- Skid Resistance: 2.7

2.5 PVC Coated Fabric

PVC-coated fabric is a type of material that consists of a woven or non-woven base fabric that has been coated with a layer of polyvinyl chloride (PVC) resin. PVC is a synthetic thermoplastic polymer that is known for its durability, flexibility, and resistance to weather, chemicals, and abrasion.

2.5.1 PVC Coated Fabric Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 4.4
- Abrasion Resistance: 4.2
- UV Resistance: 4.6
- Weather Resistance: 4.3
- Water Impermeability: 4.8
- Air Impermeability: 4.8
- Material Cost: 1.2
- Material Weight: 3.4
- Ease of Use: 4.5
- Skid Resistance: 4.5

2.6 Polyethylene Fabric

Polyethylene fabric is a thermoplastic polymer made of different crystalline structures from petroleum-based plastics.

2.6.1 Polyethylene Fabric Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 2.5
- Abrasion Resistance: 3.1
- UV Resistance: 4.5
- Weather Resistance: 4.5
- Water Impermeability: 2.1
- Air Impermeability: 1.6
- Material Cost: 1.2
- Material Weight: 2.3
- Ease of Use: 2.2
- Skid Resistance: 3.6

2.7 Polyvinal Fabric

PVC vinyl fabrics are synthetic technical fabrics engineered for use in a wide variety of applications, including sports clothing and equipment, awnings, space vehicles, fire protective clothing, automotive components, and military and commercial tents.

2.7.1 Polyvinal Fabric Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 3.9
- Abrasion Resistance: 4.3
- UV Resistance: 4.8
- Weather Resistance: 4.7
- Water Impermeability: 4.7
- Air Impermeability: 3.5
- Material Cost: 2.3
- Material Weight: 2.0
- Ease of Use: 3.8
- Skid Resistance: 2.7

2.8 Cordura Fabric

Cordura fabric is a highly durable textile known for its exceptional resistance to abrasions, tears, and scuffs. This incredible textile began as a nylon fiber developed to enhance strength and durability.

2.8.1 Cordura Fabric Material Properties

The material's following attributes will be listed below on a range from 0-5.

- Durability: 4.2
- Abrasion Resistance: 4.5
- UV Resistance: 3.8
- Weather Resistance: 3.6
- Water Impermeability: 2.6
- Air Impermeability: 2.5
- Material Cost: 1.2
- Material Weight: 1.3
- Ease of Use: 3.1
- Skid Resistance: 3.5

3 Material Ranking

To determine the most effective materials we will be performing a decision matrix. As such the values from the material attributes will be multiplied against the weights and summed together. The top three choices will be considered for final selection.

3.1 Decision Matrix

Mat.	Dbi*4.1	Abr*4.5	UV*1.3	Wthr*1.8	Wtr*4.4	Air*4.6	Cst*2.1	Wt*4.1	Ease*3.4	Skid*3.9	Tot.
NprR	4.6	4.6	4.3	4.3	2.6	3.4	4.5	2.3	3.8	1.2	116.9
NprFb	4.7	4.7	4.0	4.5	4.6	4.6	2.8	1.9	3.5	1.6	127.4
PUNyl	4.6	4.7	5.0	4.8	4.3	4.5	4.8	4.8	4.6	4.3	157.4
HypFb	4.5	4.6	4.3	4.4	5.0	4.3	3.3	3.2	4.7	2.7	141.4
PVCFb	4.4	4.2	4.6	4.3	4.8	4.8	1.2	3.4	4.5	4.5	143.6
PETFb	2.5	3.1	4.5	4.5	2.1	1.6	1.2	2.3	2.2	3.6	88.7
PVFb	3.9	4.3	4.8	4.7	4.7	3.5	2.3	2.0	3.8	2.7	123.8
CDAFb	4.2	4.5	3.8	3.6	2.6	2.5	1.2	1.3	3.1	3.5	104.2

As we can see after performing the decision matrix operation we can conclude that our top three choices are Polyurethane Coated Nylon, Hypalon Fabric, and PVC-Coated Fabric.

3.2 Final Material Selection

Here we will lay out the minute advantages that each material has over each other. Take note, all of these materials have the common trait in that they are durable, UV and weather-resistant, water-impermeable, and air-impermeable.

3.2.1 Polyurethane Coated Nylon

This material has the least weight and is the cheapest out of the three. Most polyurethane-coated nylon also have ripstock features built into them that prevent damage from propagating across the material.

3.2.2 Hypalon Fabric

This material is extremely weather-resistant. Slightly thicker which means it can take more abuse, but is also more heavy. This material is also the most durable out of the three with a 1500 psi tensile yield strength.

3.2.3 PVC-Coated Fabric

This material is the most expensive out of all three options. Has the highest impact strength and abrasive resistance out of all three materials. Its weight is slightly less than that of Hypalon fabric.

4 Fabrication Method Suggestions

There exists polyurethane coating and silicon coating canisters that may be used to coat any of the materials listed above to give them additional material properties.

As per the guidelines we are not permitted to use adhesives, but we have seen methods utilizing sewing techniques to model the necessary geometry to accomplish the hovercraft project. Such instructions will be provided separately.

5 Links

5.1 Supplier Links

5.1.1 Neoprene Rubber

https://a.co/d/5QIHV2M

5.1.2 Neoprene Fabric

https://a.co/d/ieIz3Kp

5.1.3 Polyurethane-coated Nylon

https://a.co/d/8jAG0SP

5.1.4 Hypalon Fabric

https://a.co/d/8oFCaMB

5.1.5 PVC Coated Fabric

https://a.co/d/gdS8WNR

5.1.6 Polyethylene Fabric

https://a.co/d/d7qZpf6

5.1.7 Polyvinyal Fabric

https://a.co/d/0LoEMao

5.1.8 Cordura Fabric

https://rockywoods.com/collections/cordura-nylon